News

Added to the Nanoplane series for LCD and semiconductor manufacturing equipment

November 30, 2000

シェアする ポスト

Sumitomo Heavy Industries, Ltd. (President: Yoshiro Niino) is pleased to announce that it has commercialized ultra-precision positioning equipment for liquid crystal and semiconductor equipment and added it to the "Nanoplane" series.

In order to respond to the miniaturization of devices and the improvement of throughput, the positioning equipment installed in semiconductor manufacturing equipment is required to be more precise and high-speed than ever.
The MODEL SLHUS-370 ultra-precise XY stage, which enables high-speed, nano-meter positioning accuracy at a level of one billion meters, has been commercialized.
The stop accuracy has been achieved below the industry's highest level of ±10nm (nanometers), and the positioning accuracy required for next-generation semiconductor lithography equipment and inspection equipment has been achieved. In addition, the maximum speed is 2.5 times faster than our existing model, at 300mm per second. In addition to adopting a laser interferometer, it is equipped with a control mechanism that combines a disturbance observer that improves the scanning performance of the stage and an active yaw control that suppresses yog. In addition, it uses a completely non-contact drive system with air bearings and linear motor drives, realizing high speed, high precision, cleanness, and maintenance-free.
We respond to user needs as a precision positioning equipment manufacturer with a diverse lineup of multi-axis stages such as single-axis stages to multi-axis stages such as XYZθ, which supports general-purpose stages using ball screws to ultra-precision motor-driven ultra-precise stages.

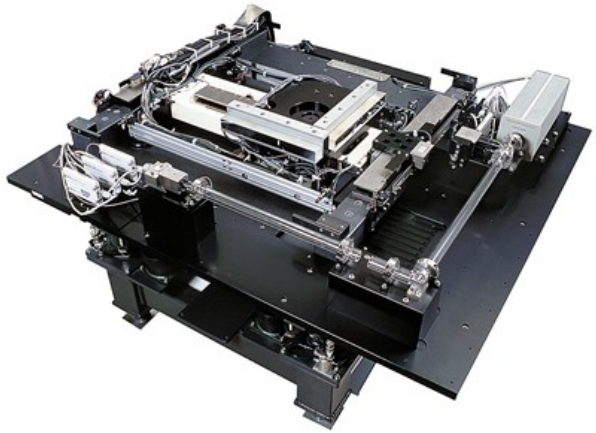
The ultra-precision positioning device "MODEL SLHUS-370" will be exhibited at the SEMICON 2000 to be held at Makuhari Messe from December 6th next week.

* "Nanoplane" is a registered trademark of Sumitomo Heavy Industries.

(3) External disturbance observer and actable y-o-control to the control mechanism
Adoption (our patent). Improved positioning accuracy and scanning accuracy.
(4) Adopts a unique H-type stage structure on the X-axis and Y-axis on the same plane.
Low center of gravity improves exercise accuracy and achieves compactness at the same time.
(5) With complete non-contact driving with linear motors and air bearings,
Cleanliness and maintenance-free are realized.

2. Main Specifications:
X-axis, Y-axis common
(1) Stroke: 370mm
(2) Scale resolution: 0.6nm (laser interferometer)
(3) Max speed: 300mm/s
(4) Stop accuracy: ±10nm or less
(5) Straightness (horizontal/vertical) 12nm/20nm (control correction value)

3. Price:
About 30million yen (depending on the specification)



Products and Solutions

Mechatronics

Drive Technologies SBU
- Gear Motor
- Large gearbox
- Deceleration gear for precision control
- Motor/Inverter
- Drive system
- AGV/AMR Drive Solution
- Autonomous mobile robot

Advanced Technologies SBU
- Cryogenic chiller
- Cryopump
- Vacuum Robot
- Precision Positioning Equipment
- Motion Components

Industrial Machinery

Material Solutions SBU
- Ion implantation equipment
- Ion Beam Irradiation Technology Service

Medical & Kantam Solutions SBU
- Cyclotron system for PET
- Proton-electron treatment system
- Electron Irradiation Service
- Vacuum deposition equipment

Metal Processing Solutions SBU
- Forged Press
- Lifting Magnets
- Coolant treatment unit

Plastic Solutions SBU
- Injection Molding Machine

Spinning Machine

Dust collector

Cooling Tower

- Laser related equipment
- PET Drug Synthesis System
- BNCT
- Diagnostics and Diagnostics Services
- STAF (Steel Tube Air Forming)
- Rolls for rolling
- Blades for aircraft jet engines
- Laminator equipment

Cleanroom System

Kneading machine

Logistics & Construction

Transport system SBU
- Carry-on cargo handling
- Logistics system

Energy & Lifeline

Energy Environment SBU
- Liquefied air energy storage (LAES)
- Circulating fluidized bed (CFB) boiler
- Waste-to-energy plants
- Electric duster
- Ashes treatment equipment
- CFB Sclava
- Digital Service (Plant Operation Support System)
- Fluidized Bed Gasification Furnace
- Bubbling Fluidized Bed (BFB) Boiler
- Wasted heat boiler
- Bug filter type exhaust gas treatment equipment
- Evaporation and crystallization equipment
- Flue gas denitrification device
- Rotary Kiln-type recycling facility

Process Plant SBU
- Stirr tank
- Plant Engineering
- Cork drum and reaction vessel
- Offshore wind power business
- Water and Pure Water Treatment Facilities
- Food Machinery
- Distillation and extraction equipment
- Coke Furnace Machine
- Steel Structures
- Sewage treatment plant
- Industrial Wastewater Treatment Facilities

Oceans and Steel Structures SBU
- Offshore wind power business
- Wind Propulsion Components

Steam turbines

Process Pump

Others

IT Solutions

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machinery (earth drill, ground improvement machine, etc.)

Link Belt Crane SBU

Mobile Crane

Construction Machinery SBU

Excavator

Road machinery

Forklift

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Sumitomo Heavy Industries Technical Report

Manufacturing of Sumitomo Heavy Industries

The front line of technology - world-class technology -

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Sumitomo Heavy Industries, Ltd.

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Sumitomo Heavy Industries (Vietnam) Co., Ltd.
Sumitomo Machinery Corporation of America
Lafert S.p.A.
Sumitomo (SHI)Cyclo Drive Germany GmbH
住友重機械減速機(中国)有限公司
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アドバンストテクノロジーズSBU

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Sumitomo(SHI)Demag Plastics Machinery GmbH
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LBX Company,LLC
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その他

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